

ASSIGNMENT – 1

Working with Python packages - NumPy, Pandas, Scikit-Learn, Matplotlib

AIM:

To explore and gain practical experience with fundamental Python libraries used in Machine Learning — NumPy, Pandas, Scikit-learn and Matplotlib — and to perform data preprocessing and exploratory data analysis (EDA) on the Diabetes Prediction dataset, covering missing value handling, irrelevant feature removal, categorical encoding, feature scaling, correlation and mutual information analysis, visualization, and dataset splitting for model development.

SOURCE CODE:

```
# ASSIGNMENT

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt

#Load the dataset into Python using the Pandas library.

df = pd.read_csv("Diabetes_prediction(2).csv")
print("Dataset loaded succesfully\n")

#Display the first five records of the dataset.

print("displaying first five record")
print(df.head())

#Determine: Number of rows, Number of columns, Feature names, Target (label) column

print("Number of rows\n")
print(df.shape[0])

print("Number of columns\n")
print(df.shape[1])

print("Feature Name\n")
print(df.columns.tolist()) #to list to display it has list

print("Target(label) Column\n")
print(df.columns[-1])
```

```
#Data Preprocessing:

#Identify and handle missing values- Explain the method used to handle missing values.

print("Missing values")
print(df.isnull().sum()) #before

df.fillna(df.mean(numeric_only=True),inplace=True)
#or
# df = df.fillna(df.mean(numeric_only=True))

print(df.isnull().sum()) #after


#Remove irrelevant features - Justify whether they should be removed.

if "id" in df.columns:
df.drop("id",axis=1,inplace=True)
print(" ID column removed\n")
else:
print("No irrelevent feture found\n")


print("Duplicates Before Removal :", df.duplicated().sum())

df = df.drop_duplicates()

print("Duplicates After Removal :", df.duplicated().sum())


# Encode categorical variables - Justify the technique used for encoding

from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder() #object


categorical = df.select_dtypes(include="object").columns

for col in categorical:

df[col] = encoder.fit_transform(df[col])

print("\nCategorical Features Encoded")
```

```
from sklearn.preprocessing import StandardScaler

# Create scaler object
scaler = StandardScaler()

# Only scale input features
numerical_features = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age"
]

#the target feature should not be mentioned here

df[numerical_features] = scaler.fit_transform(df[numerical_features])

print(df.head())

# Correlation Matrix
correlation = df.corr(numeric_only=True)

print(correlation)

print("\nCorrelation with Target Variable:")
print(correlation["Diagnosis"].sort_values(ascending=False))

#Mutual Information
from sklearn.feature_selection import mutual_info_classif
X = df.drop("Diagnosis",axis=1)
y = df["Diagnosis"]

mi = mutual_info_classif(X,y)
mi_scores = pd.Series(mi,index=X.columns)
print("Mutual Information Scores")
print(mi_scores.sort_values(ascending=False))

#Important Features
importance = pd.DataFrame({
```

```
"Feature": X.columns,
"Correlation": correlation["Diagnosis"].drop("Diagnosis").values,
"Mutual Information": mi_scores.values
})

print("Important Features")
print(importance)

print(mi_scores.head(10))

#Exploratory Data Analysis (EDA) and Visualization:

#Summary statistics
print("Summary statistics")
print(df.describe())

#Histograms,
df.hist(figsize=(14,10))
plt.suptitle("Histogram of Fetures")
plt.show()

#Bar Chart
# class_count = df["Diagnosis"].value_counts()

# plt.figure(figsize=(7,5))

# plt.bar(
# class_count.index.astype(str),
# class_count.values,
# color=["skyblue", "darkblue"]
# )

# plt.xlabel("Diagnosis")
# plt.ylabel("Count")
# plt.title("Target Distribution")

# plt.show()

class_count = df["Diagnosis"].value_counts()

plt.figure(figsize=(7,5))

plt.bar(
class_count.index.astype(str),
class_count.values,
```

```
color=["skyblue","orange"]
)
plt.xlabel("Diagnosis")
plt.ylabel("count")
plt.title("Target Distribution")

plt.show()

#Scatter Plot

plt.figure(figsize=(7,5))

diabetic = df[df["Diagnosis"]==1]
non_diabetic = df[df["Diagnosis"]==0]

plt.scatter(non_diabetic["Glucose"],
non_diabetic["BMI"],
color="blue",
label="Non-Diabetic")

plt.scatter(diabetic["Glucose"],
diabetic["BMI"],
color="red",
label="Diabetic")

plt.xlabel("Glucose")
plt.ylabel("BMI")
plt.title("Glucose vs BMI")

plt.legend()

plt.show()

# Box Plot

plt.figure(figsize=(12,6))

numeric_columns = df.select_dtypes(include=np.number).columns

plt.boxplot([df[col] for col in numeric_columns],
tick_labels=numeric_columns)

plt.xticks(rotation=45)

plt.title("Box Plot of Numerical Features")
```

```
plt.show()

#Heatmap

corr = df.corr(numeric_only=True)

plt.figure(figsize=(10,8))

plt.imshow(corr, cmap="coolwarm")

plt.colorbar()

plt.xticks(range(len(corr.columns)),
corr.columns,
rotation=90)

plt.yticks(range(len(corr.columns)),
corr.columns)

plt.title("Correlation Heatmap")

plt.show()

#Pair Plot

plt.figure(figsize=(6,5))

plt.scatter(df["Glucose"],
df["BMI"])

plt.xlabel("Glucose")
plt.ylabel("BMI")
plt.title("Glucose vs BMI")

plt.show()

# Class Balance

print("\nClass Distribution")

count = df["Diagnosis"].value_counts()

plt.figure(figsize=(6,5))

plt.bar(count.index.astype(str),
count.values,
```

```
color=["green","red"])

plt.xlabel("Diagnosis")
plt.ylabel("Number of Samples")
plt.title("Class Distribution")

plt.show()

# Data Splitting

from sklearn.model_selection import train_test_split

X_train, X_temp, y_train, y_temp = train_test_split(

X,

y,

test_size=0.30,

random_state=42

)

X_validation, X_test, y_validation, y_test = train_test_split(

X_temp,

y_temp,

test_size=0.50,

random_state=42

)

print("\nTraining Shape")

print(X_train.shape)

print("\nValidation Shape")

print(X_validation.shape)

print("\nTesting Shape")

print(X_test.shape)
```

OUTPUT:

Dataset loaded succesfully

displaying first five record

Pregnancies Glucose ... Age Diagnosis

0 2 115.863387 ... 20.100494 0

1 2 92.490122 ... 44.912281 0

2 1 88.141469 ... 48.247873 1

3 2 108.453101 ... 42.749868 0

4 1 127.849443 ... 32.797789 0

[5 rows x 9 columns]

Number of rows

1000

Number of columns

9

Feature Name

['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI',
'DiabetesPedigreeFunction', 'Age', 'Diagnosis']

Target(label) Column

Diagnosis

Missing values

Pregnancies 0

Glucose 0

BloodPressure 0

SkinThickness 0

Insulin 0

BMI 0

DiabetesPedigreeFunction 0

Age 0

Diagnosis 0

dtype: int64

Pregnancies 0

Glucose 0

BloodPressure 0

SkinThickness 0

Insulin 0

BMI 0

DiabetesPedigreeFunction 0

Age 0

Diagnosis 0

dtype: int64

No irrelevant feature found

Duplicates Before Removal : 0

Duplicates After Removal : 0

Categorical Features Encoded

Pregnancies Glucose ... Age Diagnosis

0 0.169163 0.843882 ... -1.603337 0

1 0.169163 -0.357150 ... 0.112773 0

2 -0.569542 -0.580604 ... 0.343479 1

3 0.169163 0.463106 ... -0.036791 0

4 -0.569542 1.459784 ... -0.725127 0

[5 rows x 9 columns]

Pregnancies Glucose ... Age Diagnosis

Pregnancies 1.000000 0.011362 ... 0.088453 -0.001484

Glucose 0.011362 1.000000 ... -0.003056 -0.004971

BloodPressure 0.077683 0.010261 ... 0.049709 0.010349

SkinThickness -0.040438 0.002874 ... -0.047309 -0.000014

Insulin -0.020708 -0.010334 ... 0.037062 0.035411

BMI -0.006228 0.002024 ... -0.066314 -0.033336

DiabetesPedigreeFunction -0.001419 0.005122 ... 0.017163 0.060549

Age 0.088453 -0.003056 ... 1.000000 -0.044726

Diagnosis -0.001484 -0.004971 ... -0.044726 1.000000

[9 rows x 9 columns]

Correlation with Target Variable:

Diagnosis 1.000000

DiabetesPedigreeFunction 0.060549

Insulin 0.035411

BloodPressure 0.010349

SkinThickness -0.000014

Pregnancies -0.001484

Glucose -0.004971

BMI -0.033336

Age -0.044726

Name: Diagnosis, dtype: float64

Mutual Information Scores

DiabetesPedigreeFunction 0.008826

Insulin 0.002899

Pregnancies 0.002453

Age 0.002413

BMI 0.001744

SkinThickness 0.000000

Glucose 0.000000

BloodPressure 0.000000

dtype: float64

Important Features

Feature Correlation Mutual Information

0 Pregnancies -0.001484 0.002453

1 Glucose -0.004971 0.000000

```
2 BloodPressure 0.010349 0.000000
3 SkinThickness -0.000014 0.000000
4 Insulin 0.035411 0.002899
5 BMI -0.033336 0.001744
6 DiabetesPedigreeFunction 0.060549 0.008826
7 Age -0.044726 0.002413
Pregnancies 0.002453
Glucose 0.000000
BloodPressure 0.000000
SkinThickness 0.000000
Insulin 0.002899
BMI 0.001744
DiabetesPedigreeFunction 0.008826
Age 0.002413
dtype: float64
Summary statistics
Pregnancies Glucose ... Age Diagnosis
count 1.000000e+03 1.000000e+03 ... 1.000000e+03 1000.00000
mean 1.136868e-16 -3.090861e-16 ... -1.065814e-17 0.30600
std 1.000500e+00 1.000500e+00 ... 1.000500e+00 0.46106
min -1.308246e+00 -3.538833e+00 ... -3.061357e+00 0.00000
25% -5.695415e-01 -6.831450e-01 ... -6.752827e-01 0.00000
50% 1.691634e-01 9.123268e-04 ... 2.437897e-02 0.00000
75% 9.078684e-01 7.103414e-01 ... 6.789692e-01 1.00000
max 4.601393e+00 3.175497e+00 ... 3.270954e+00 1.00000
```

```
[8 rows x 9 columns]
```

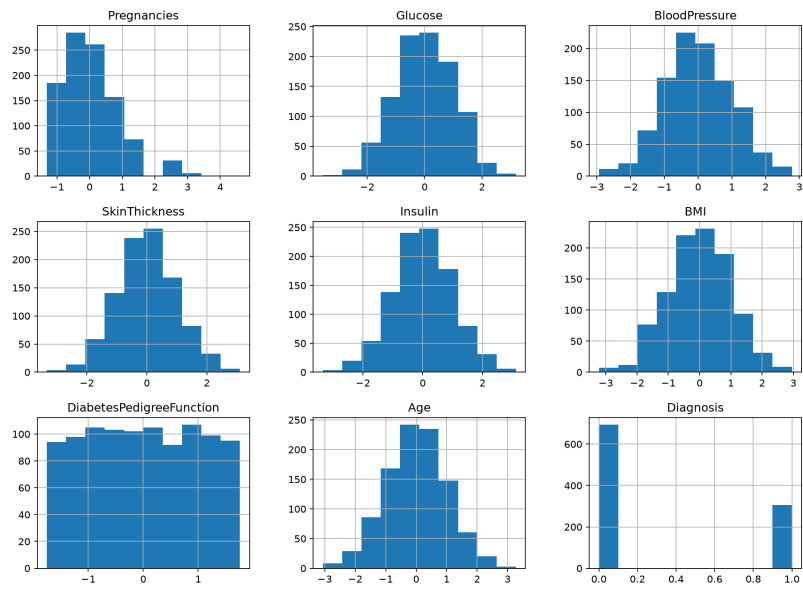
```
Class Distribution
```

```
Training Shape
(700, 8)
```

```
Validation Shape
(150, 8)
```

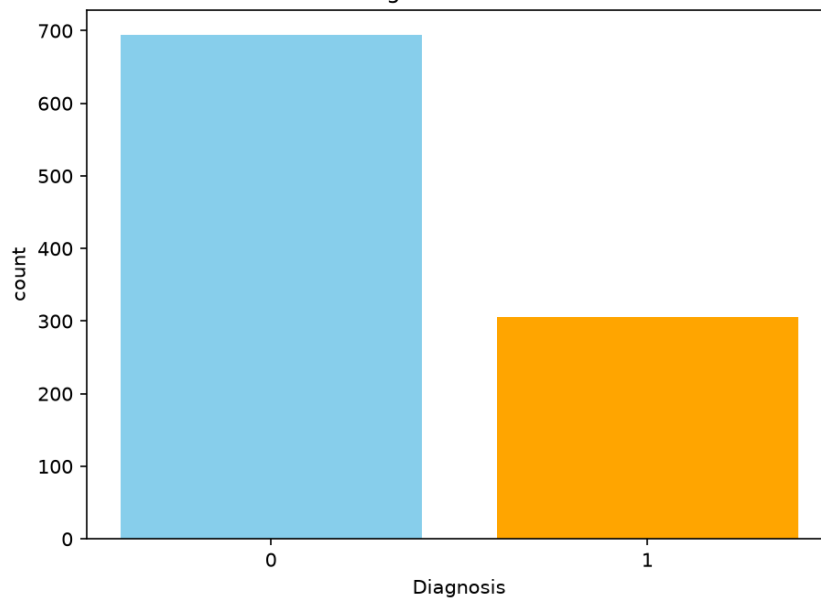
```
Testing Shape
(150, 8)
```

Histogram of Fetures

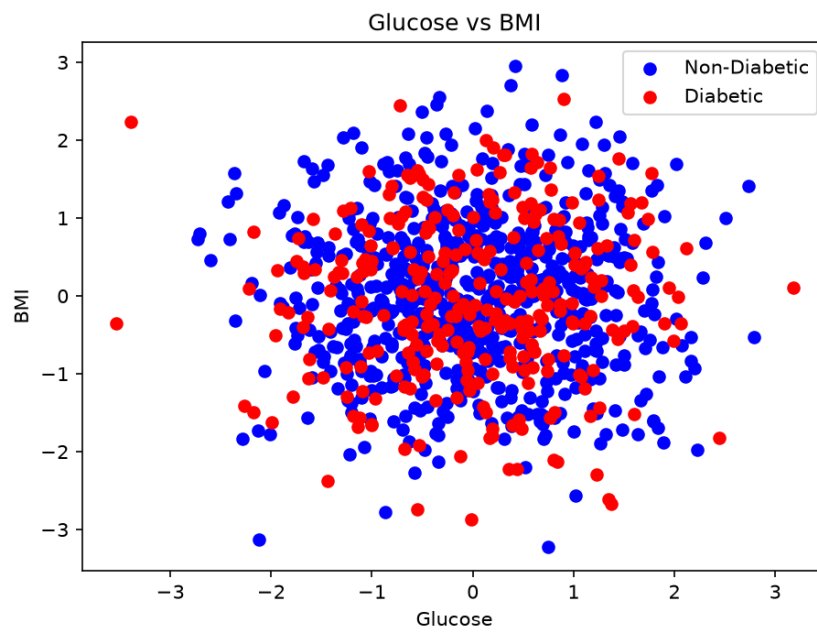


Histogram of Features

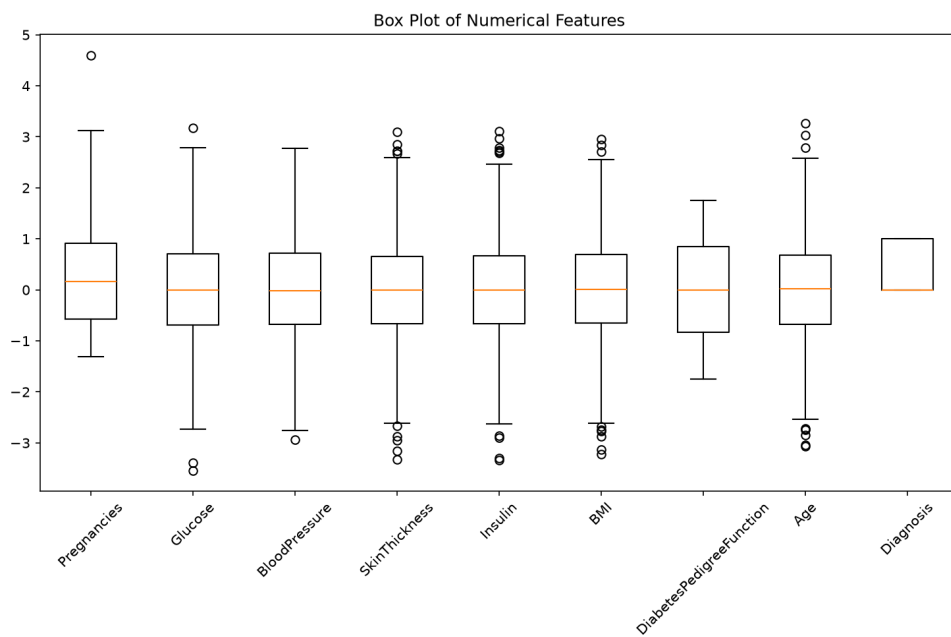
Target Distribution



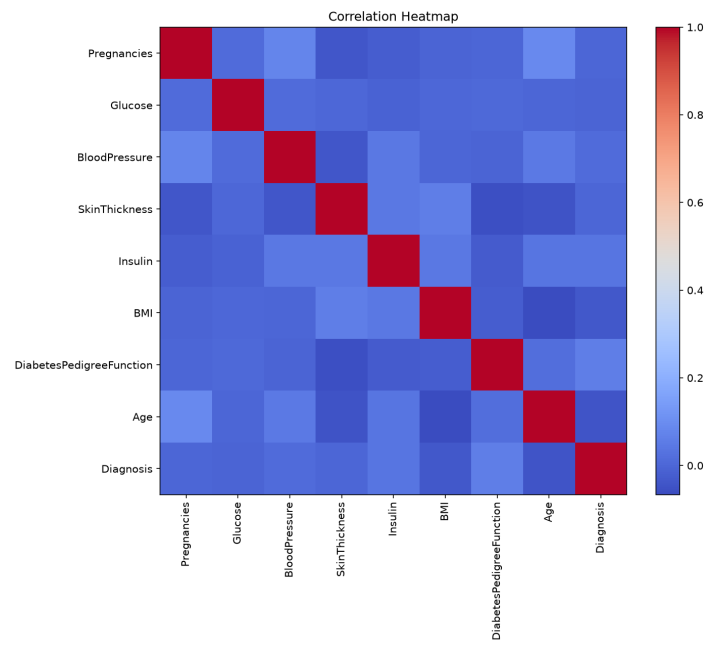
Bar Chart - Target Distribution



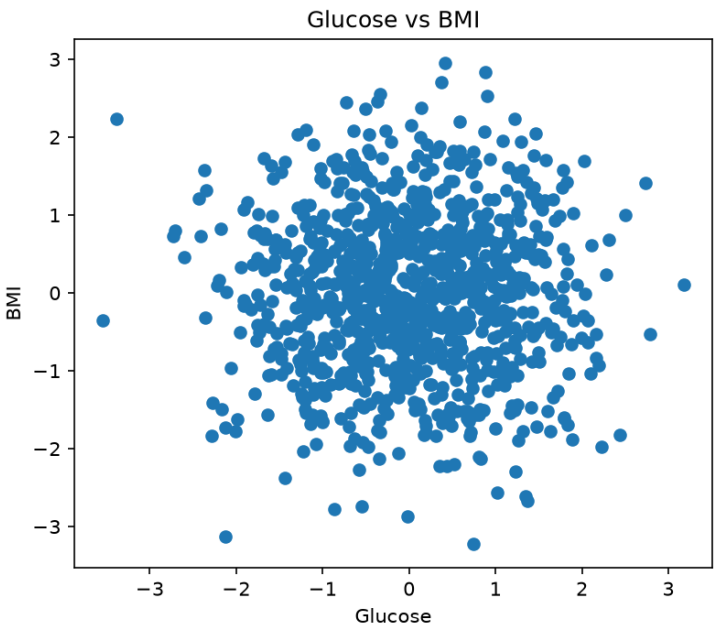
Scatter Plot - Glucose vs BMI (Diabetic / Non-Diabetic)



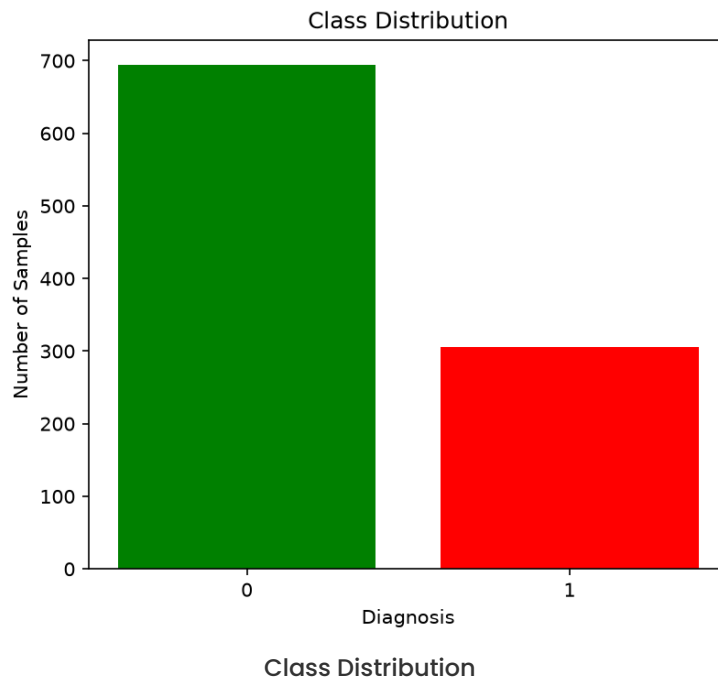
Box Plot of Numerical Features



Correlation Heatmap



Pair Plot - Glucose vs BMI



Tasks Performed

S.No	Task Performed	Observations / Findings
1	Dataset Loading	Diabetes Prediction dataset (1000 rows, 9 columns) loaded successfully into a Pandas DataFrame.
2	First Five Records	Verified dataset structure; confirmed correct loading of all 9 feature columns.
3	Dataset Information	Identified 1000 rows, 9 columns, 8 input features, and target column 'Diagnosis'.
4	Missing Value Handling	No missing values found in any column; mean imputation was applied as a precaution.
5	Duplicate Removal	No duplicate rows found (0 before and after removal).
6	Irrelevant Feature Check	No 'id' column present, so no irrelevant feature was removed.
7	Categorical Encoding	Dataset contained no categorical (object-type) columns, so no encoding was required.
8	Feature Scaling	All 8 numerical input features standardized using StandardScaler (mean 0, std 1).
9	Correlation Analysis	DiabetesPedigreeFunction (0.061) and Insulin (0.035) showed the strongest, though weak, positive correlation with Diagnosis.
10	Mutual Information Analysis	DiabetesPedigreeFunction was the most informative feature, followed by Insulin and Age.

S.No	Task Performed	Observations / Findings
11	Summary Statistics	Generated descriptive statistics (mean, std, min, max, quartiles) for all features.
12	Histogram	Most features approximately follow a normal distribution after scaling; Diagnosis is imbalanced.
13	Bar Chart	Visualized target distribution: 694 non-diabetic vs 306 diabetic samples.
14	Scatter Plot	Glucose vs BMI shows overlapping clusters for diabetic and non-diabetic samples, no clear separation.
15	Box Plot	Identified potential outliers in Insulin and Pregnancies.
16	Heatmap	No strong pairwise correlations observed among the numerical features.
17	Pair Plot	Confirmed no obvious linear relationship between Glucose and BMI.
18	Class Balance	Dataset is imbalanced: 69.4% non-diabetic, 30.6% diabetic.
19	Data Splitting	Dataset split into Training (700), Validation (150), and Testing (150) sets.

Learning Outcomes

After completing this experiment, I was able to:

- Understand the purpose and applications of the Python libraries NumPy, Pandas, Scikit-learn, and Matplotlib.
- Perform data loading, indexing, filtering, and statistical analysis using Pandas.
- Handle missing values, remove irrelevant features, and detect duplicate records as part of data cleaning.
- Encode categorical variables and scale numerical features using Scikit-learn preprocessing tools.
- Analyze feature relevance using correlation and mutual information techniques.
- Visualize data distributions and relationships using histograms, bar charts, scatter plots, box plots, and heatmaps.
- Split a dataset into training, validation, and testing sets for model development.